0:00

0 seconds

All right, let's do some dosage calc.

0:03

3 seconds

Um, this one we're going to be talking about way weightbased medications and

0:11

11 seconds

most of these will involve kids whether we're giving them oral medications, IV medications,

0:20

20 seconds

that doesn't matter. in kiddos because their sizes are so different. Um your medications for kiddos will be weightbased.

0:30

30 seconds

Um there are a few medications for adults for my non-pedes people. um that we do

0:38

38 seconds

incorporate weightbased dosing and those medications that I can think of off the top of my head are Hepin, Levoped

0:49

49 seconds

um I don't know those are the only ones that are coming to me but a lot of Oh,

0:57

57 seconds

and like IV fluids for um sepsis workups or um sepsis management

1:07

1 minute, 7 seconds

uh fluid resuscitation or IV fluids for um volume replacement, whatever. Doesn't

1:15

1 minute, 15 seconds

matter. You guys don't care. Anyways, we do use some weight- based dosing for adults, but the majority of this you're

1:23

1 minute, 23 seconds

going to find in kids. So we are used to seeing two units here in our problems

1:32

1 minute, 32 seconds

and now these problems we're going to have three units. So we're going to have something like 20 MIGs per kig per day.

1:40

1 minute, 40 seconds

And I will show you a couple different ways to solve these problems.

1:44

1 minute, 44 seconds

Okay. So how the order is for 20 milligrams per kilogram per day. The

1:52

1 minute, 52 seconds

patient weighs 112 pounds. How many milligrams can they have per day? So, you can do it the, you know, common dimensional analysis

2:01

2 minutes, 1 second

way. So, I always like to start out with what is being asked of me. Okay, at the end I should have milligrams per day.

2:09

2 minutes, 9 seconds

So, I'm going to set up my problem and I'm going to say, okay, I always just kind of start with what's on top here because it's just easiest for my brain

2:17

2 minutes, 17 seconds

to function that way. So, I plug in my milligrams. Okay, I know it's 20 milligrams per kilogram. All right, then

2:24

2 minutes, 24 seconds

I got to get rid of kilogram. So, it's 1 kilogram is 2.2 lb and the patient weighs 112

2:31

2 minutes, 31 seconds

lbs. And a lot of people get here and go, "Well, I don't know what else to put." Well, here's where your extra unit will come in and it just goes under your pounds because that's all we have left.

2:40

2 minutes, 40 seconds

Okay, your kilos will cross out, your pounds will cross out. You're left with milligrams over day.

2:47

2 minutes, 47 seconds

And because I Oh, there it is. Like because I am

2:54

2 minutes, 54 seconds

a dumb dum and I didn't solve these beforehand.

3:00

3 minutes

I'm going to solve it right now. And your answer will be a,000 18.1818.

3:10

3 minutes, 10 seconds

All right. If we were rounding to the nearest whole number, it would be 1,8.

3:17

3 minutes, 17 seconds

If we rounding to the nearest tenth, it would be 118.2. Okay.

3:25

3 minutes, 25 seconds

Where this is going to Oh, you know what? Sorry. Let me show you the other way that uh you can do this. Okay.

3:33

3 minutes, 33 seconds

Or at least this is the other way that kind of makes sense in my brain. Okay. There are like

3:41

3 minutes, 41 seconds

lots of different ways you can do these problems. So, I'm looking for milligrams per day. All right, that's what I want.

3:49

3 minutes, 49 seconds

If you if this like three unit thing freaks you out, just get rid of it.

3:53

3 minutes, 53 seconds

Okay, so the first thing I I read this as well and how you should read it is 20

3:59

3 minutes, 59 seconds

milligrams per kilogram per day aka per

4:07

4 minutes, 7 seconds

is always multiplied. Okay. So, I could first get rid of this like 112

4:14

4 minutes, 14 seconds

pound. I can I can look at 20 migs per kig as one thing and then you know when I figured out the migs per k I'm just left with migs per day.

4:26

4 minutes, 26 seconds

Anyways, it'll make sense. I hope. Okay.

4:29

4 minutes, 29 seconds

So, I can first convert my 112 pounds divided by 2.2 2 and that is so 112

4:36

4 minutes, 36 seconds

pounds is 50.999 kg. Okay. And then I can just multiply

4:46

4 minutes, 46 seconds

that times 20 millig because it's 20 millig per kilogram. So if I have 50

4:52

4 minutes, 52 seconds

kilos times 20, okay, that is 18.1818.

5:02

5 minutes, 2 seconds

And I know that's per day. So I'm left with milligrams 1,8 milligrams per day

5:11

5 minutes, 11 seconds

because that's the only unit I really have left. Um well or you know since I told you it was

5:20

5 minutes, 20 seconds

this is kind of like multiplied the per if you read it as per that's kind of multiplication.

5:26

5 minutes, 26 seconds

So then it becomes 118.1818 times one day. Okay, if this was order

5:33

5 minutes, 33 seconds

states administer 20 milligrams per kilo per three days, then it would be 118 times three and that would if it says

5:43

5 minutes, 43 seconds

per 3 days, then that's how you get your answer. Okay.

5:49

5 minutes, 49 seconds

So, um, there's plenty of these um, examples

5:59

5 minutes, 59 seconds

where it's going to be different. When you have these three units and they're going to ask you different things,

6:06

6 minutes, 6 seconds

they're either going to ask you per day or they're going to ask you per dose.

6:12

6 minutes, 12 seconds

And if they're going to ask you per dose, they have to tell you how. Um, no, sorry, they don't have to tell you that.

6:18

6 minutes, 18 seconds

If they're going to ask you per day, they either have to specify day

6:25

6 minutes, 25 seconds

or if they don't give you that, they either have to that they have to say something like, "Oh, the patient's going

6:33

6 minutes, 33 seconds

to get it every eight hours." Okay, so I show you what I mean.

6:40

6 minutes, 40 seconds

All right, let's do this one. So, tobery four MIGs per kick per day to be administered every eight hours.

6:49

6 minutes, 49 seconds

The patient weighs 60 kilos and you have this available. That's your available concentration. So, how many mills should

6:56

6 minutes, 56 seconds

the nurse give per dose? Okay, I'll do it the dimensional analysis way. I'm going to start with 2 mills because

7:04

7 minutes, 4 seconds

that's what I have on top. I'm just going to use my mill.

7:09

7 minutes, 9 seconds

40 milligrams, four milligrams per kilo. I've got 60 kilos and I'm left

7:18

7 minutes, 18 seconds

with per day. Okay, but that's not asking day this time. If it said per day, then I would leave it.

7:27

7 minutes, 27 seconds

But they're asking per dose. So they are giving me a day. And so now I have to think in one day, how many doses is this

7:36

7 minutes, 36 seconds

patient going to get? Well, in one day they get it every eight hours.

7:42

7 minutes, 42 seconds

So 24 divided by 8 is three. So they are going to have three doses

7:49

7 minutes, 49 seconds

in one day. Okay. And then my units will cross out and I am left with let me calculate 2 *

7:59

7 minutes, 59 seconds

4 * 60 / 40 / 3 is 4. So four mills per dose. If we

8:09

8 minutes, 9 seconds

were to do mills per day, we would have stopped here. Okay. So

8:17

8 minutes, 17 seconds

your answer is 12 for mills per day because then we don't have to convert our day like how many doses do we have in a day.

8:24

8 minutes, 24 seconds

Okay. And that is how we do those. Let's do this one. All right.

8:32

8 minutes, 32 seconds

How many mills will the nurse administer? And this is giving me this says administer laurazzipam.05

8:40

8 minutes, 40 seconds

migs per kig one time before surgery. So if it's asking me how many mills I'm going to give, it's for one time. So per dose.

8:49

8 minutes, 49 seconds

Okay.

8:52

8 minutes, 52 seconds

All right. So let's plug in my numbers here. How many mills? So I've got one mil 2 milligrams.

8:59

8 minutes, 59 seconds

I've got 0.05 milligrams per kilogram.

9:04

9 minutes, 4 seconds

I've got 1 kilogram is 2.2 pounds and I've got 192 pounds

9:12

9 minutes, 12 seconds

and that's going to be per dose because that's all I have left. I used everything else. So migs cross out, ks

9:18

9 minutes, 18 seconds

cross out, pounds cross out and I've got 005 * 192

9:27

9 minutes, 27 seconds

/ 2 / 2.2 two and this is 2.1818.

9:35

9 minutes, 35 seconds

All right. So round to the nearest whole number two. Okay.

9:42

9 minutes, 42 seconds

Weight-based practice. Here we go. You have on hand delotted 4 milligrams per mill. You need to administer 0.015 migs

9:50

9 minutes, 50 seconds

per kg to a patient who weighs 150 pounds. How many milliliters should you administer per dose?

9:57

9 minutes, 57 seconds

Okay, this is asking per dose because I didn't have how many doses I have a day, how often, or a day. So, I'm going to

10:05

10 minutes, 5 seconds

assume dose here. Okay, so I'm looking for milliliters per dose.

10:12

10 minutes, 12 seconds

And that would be 1 mil and 4 milligrams times 0. Oh, sorry. Oh, yeah. 015 milligrams per kilogram.

10:26

10 minutes, 26 seconds

1 kilogram is 2.2 pounds and my patient is 150 pounds and that's dose.

10:34

10 minutes, 34 seconds

Okay, everything crosses out.

10:38

10 minutes, 38 seconds

I am left with 0.015 * 150 / 4 / 2.2.

10:49

10 minutes, 49 seconds

All right, this is 0.25568. two, five, five, six,

10:55

10 minutes, 55 seconds

eight. All right. So, round I if I write this down and then I'm like, okay, what's it asking me to round to? I

11:02

11 minutes, 2 seconds

always go at least four decimal places out so that you don't get a rounding error. Okay. Um, so two decimal places would be 0.2.

11:14

11 minutes, 14 seconds

You look at the five next door, that's up. I'm going to round up to six. So 26 milliliters per dose.

11:23

11 minutes, 23 seconds

Okay.

11:31

11 minutes, 31 seconds

All right. Things that need to be in a in an order, a medication order. Um when

11:38

11 minutes, 38 seconds

a doctor writes an order or aka prescription, a prescription is a medication order. Okay.

11:47

11 minutes, 47 seconds

Um, some of the things that need to, well, sorry, not some of the things, all of the things that need to be included in that are the patient's name, the date

11:54

11 minutes, 54 seconds

and time the prescription was written, the name of the medication, the reason for the medication, especially if it is a PRN medication

12:02

12 minutes, 2 seconds

because those PRN medications, and for you to practice within your scope of practice,

12:10

12 minutes, 10 seconds

we do not practice medicine. We practice nursing. So the doctor needs to be very specific as to why we are giving a

12:19

12 minutes, 19 seconds

medication and so they should be very specific. So PRN pain one to four or pain scale one to four. Okay.

12:32

12 minutes, 32 seconds

They need to be very specific as to when we should be giving these. The dose. So, the amount, the frequency, the strength,

12:42

12 minutes, 42 seconds

the route, specific instructions, and the physician's signature.

12:48

12 minutes, 48 seconds

There are a lot of abbreviations to avoid. Um, I think I have it pulled.

12:54

12 minutes, 54 seconds

Nope, I don't. Okay. There's lots of abbreviations to avoid, though. Ones we're pointing out here is QD. All

13:02

13 minutes, 2 seconds

right. This is supposed to be once daily, but it looks like QID, which is four times a day. So, we don't want to

13:09

13 minutes, 9 seconds

use QD. HS could be half strength in terms of medications. It could mean before bed.

13:17

13 minutes, 17 seconds

DC could mean discharge or discontinue.

13:20

13 minutes, 20 seconds

Okay, so any of these like vague abbreviations, we do not want to use them. Um, I need

13:26

13 minutes, 26 seconds

to find because I actually have a good list in here.

13:32

13 minutes, 32 seconds

Um here we go. Okay.

13:37

13 minutes, 37 seconds

Um Institute for Safe Medication Practices.

13:42

13 minutes, 42 seconds

All right. These are all the abbreviations that we do not use.

13:46

13 minutes, 46 seconds

Um and it will tell you like what's the intended meaning and what's the misunder interpretation for why we don't use it.

13:55

13 minutes, 55 seconds

Um, so we should if we see a an order like a handwritten order with

14:03

14 minutes, 3 seconds

one of these abbreviations in it, we need to be calling that physician to clarify because maybe they meant HS as

14:11

14 minutes, 11 seconds

in half strength, but we know HS as bedtime. So is do you want a half strength or do you want it at bedtime?

14:19

14 minutes, 19 seconds

we need to be calling if they're if our physicians are using these abbreviations. Okay, the ones we pointed

14:26

14 minutes, 26 seconds

out are here. So, same thing. If a prescription is missing a piece of one or the

14:36

14 minutes, 36 seconds

oh, excuse me, more one or more of the following criteria, then we need to be calling that physician to clarify.

14:45

14 minutes, 45 seconds

This is reading a medication label. Um, these do take a little bit of time and practice just to get used to how you read them. But I'll walk you through

14:54

14 minutes, 54 seconds

this one. Um, the first thing to note is this is the name of the medication. Okay, this is

15:01

15 minutes, 1 second

the brand name and this is the generic name. Okay, this will tell you as it is

15:09

15 minutes, 9 seconds

indicating the route. So, this is an oral solution. Can I give oral solutions IV? No, I cannot. This is oral. So this is PO. So this is going to be a liquid.

15:19

15 minutes, 19 seconds

All right.

15:21

15 minutes, 21 seconds

The concentration of the medication is 350 mg to 5 ms. So every 5 milliliters

15:29

15 minutes, 29 seconds

that you draw up, there's a 350 milligrams in that. Okay.

15:34

15 minutes, 34 seconds

It will tell you as well these oral solutions that usually it's like a powder that it comes with and you have

15:43

15 minutes, 43 seconds

you have to um reconstitute or put the solution. It's usually water in it so

15:52

15 minutes, 52 seconds

that it comes out to the appropriate concentration.

15:55

15 minutes, 55 seconds

It will tell you how many milliliters to reconstitute with or to put in the powder so that we can make the

16:03

16 minutes, 3 seconds

appropriate concentration. So that we'll see if it's a liquid that you have to mix like or comes in a powder and you

16:11

16 minutes, 11 seconds

have to add the liquid so that it's the right you know solution. It will tell you on the label the reconstitution

16:20

16 minutes, 20 seconds

instructions. So it says to prepare add 80 milliliters of water to the bottle with the powder. That's what they mean.

16:27

16 minutes, 27 seconds

And shake thoroughly. This prepares 100 ms oral solution total. Okay. So

16:35

16 minutes, 35 seconds

there's 20 doses of this medication in here. If there's 100 ms, right? I shouldn't say it that

16:44

16 minutes, 44 seconds

definitively. If I were to give this specific concentration if the patient were to take five mills of it, we have a 100 mills in the bottle. So, there's 20

16:52

16 minutes, 52 seconds

doses of it. Okay, that's more accurate of what I was trying to say. Your lot number, your expiration are here. I love

16:59

16 minutes, 59 seconds

that it expires in 2080. That's fantastic. Okay. Um, and it will tell you

17:08

17 minutes, 8 seconds

again your total milliliters in the bottle will equal 100 even though you're only

17:15

17 minutes, 15 seconds

adding 80. Okay? But the powder accounts for 20 mls, okay? Because there's probably a lot of powder in that. Okay?

17:24

17 minutes, 24 seconds

So, that's how you read a medication label. There's one more. This is zithramax.

17:30

17 minutes, 30 seconds

All right? A zithramy. So, this is your brand name. And this is your generic a zithramyin for injection for IV use only. That's your route. This is IV.

17:39

17 minutes, 39 seconds

Okay. In one vial there's 500 milligrams. Okay.

17:47

17 minutes, 47 seconds

Um it says to yield 100 milligrams per milliliter solution when reconstituted as directed.

17:57

17 minutes, 57 seconds

So this doesn't have the reconstitution directions like that. I can see. Um, but

18:04

18 minutes, 4 seconds

if you're at the end you would have 100 milligrams per milliliter and your vial has 500

18:13

18 minutes, 13 seconds

milligrams, it's safe to say that your vial has five ms. Okay. Um,

18:20

18 minutes, 20 seconds

so that's just a little bit about reading medication um, boxes, packages.

18:32

18 minutes, 32 seconds

Okay, that's all the things that they mean.